



High-Level Solution Design

PROTOTYPE PHASE HLSD

Solution Name:

Policy AI Lab

Version:

1.2

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Document Information

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Revision History

Version		Author	Summary of Changes
1.0	12/05/2026	Joey Shao	Initial draft
1.1	20/05/2026	Joey Shao	Revised based on solution owner feedback
1.2	26/07/2026	Joey Shao	Restructured V1.1 into the Prototype-phase HLSD (per phase template) and closed the OT HLSD review conditions: G-01 / G-02 — all 14 AI-Strategy agents dispositioned to capabilities and requirements (§4.0 traceability; BR-19–BR-22); G-04 — In-Scope / Out-of-Scope aligned strictly to BRD §6.1 / §6.2; G-07–G-09 — strategic KPIs (cost savings, revenue, productivity), Vision 2030 alignment and target rationale added (§5); G-10 — phased MVP delivery (§6.1). Detailed capability design, architecture, AI approach, data, risk, infrastructure and assumptions relocated to the Alpha- and Beta-phase HLSDs. Added Additional resources (DOC-01 / DOC-02).

HLSD Review Checklist

All mandatory items in red must be confirmed before submission to the Steering Group. Tick marks indicate which stakeholders should review each item.

✓	Review Item	Sec.	OT	SO	AC	CAIT	SGG	SH	CAIC	ARB
Strategy & Business										
☐	Business problem articulated with measurable impact	<u>2.1</u>	✓	✓	✓	✓				
☐	Strategic alignment to NHCI, MOMAH SSO, and AI strategy	<u>2.2</u>	✓	✓	✓	✓				
☐	Business case metrics defined with clear assumptions	<u>Exec.</u>	✓	✓	✓	✓				
☐	Existing solution landscape documented	<u>2.5</u>	✓	✓	✓	✓				
☐	Key requirements identified	<u>4</u>	✓	✓	✓	✓				
☐	Success metrics (i.e., KPIs) with baseline and targets	<u>5</u>	✓	✓	✓	✓				
☐	Executive summary self-contained	<u>Exec.</u>	✓			✓	✓	✓	✓	
Solution Design										
☐	Solution description accessible to non-technical audience	<u>3.1</u>	✓	✓	✓	✓		✓		
☐	Personas, capabilities, and user journeys defined and linked	<u>3.2–3.3–3.4</u>	✓	✓		✓				
☐	Productization potential assessed (if applicable)	<u>N/A (Alpha)</u>	✓	✓		✓				
☐	Roadmap covers all phases; MVP scope defined	<u>6.1</u>	✓	✓	✓	✓				
☐	Execution risks identified with mitigation strategies	<u>N/A (Beta)</u>	✓		✓	✓				
Governance										
☐	Stakeholders identified with responsibilities and contacts	<u>2.3</u>	✓	✓	✓	✓				
☐	Approval sign-off table complete	<u>App. B</u>	✓			✓				

OT = Oversight Team | SO = Solution Owner | AC = AI Champion | CAIT = Central AI Team | SGG = AI Steering Group | SH = Sector Head | CAIC = Central AI Committee | ARB = NHCI ARB

Note: Other bodies might be involved during the development of the HLSD (e.g. DT, MOMAH EA). Section 1 (Introduction) must be completed prior to Stage-Gate 1 but is not subject to formal review at the gate. All other sections not listed above are not mandatory for Stage-Gate 1 approval. Both will be reviewed by the OT for completeness and alignment.

In red are the mandatory section to be completed and approved by stage-gate 1

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Executive Summary

The Policy AI Lab is a high-performance decision-support platform designed to transform MOMAH's policy lifecycle from a fragmented, document-centric, and reactive process into a unified, AI-augmented, predictive, and evidence-based decision-intelligence capability. By integrating advanced simulation, generative AI, conflict detection, and a canonical policy registry, the platform ensures that policies requiring simulation under approved trigger rules are stress-tested against Vision 2030 KPIs and fiscal responsibility criteria, while all policies complete required conflict, legal, and governance checks before enactment.

Solution Purpose

The platform provides MOMAH Policy Analysts, Housing Economists, Legal Officers, Senior Leaders, and the Minister with role-based AI tools to draft, simulate, benchmark, and approve policies within a single governed environment. It directly addresses the root cause of MOMAH's current policy challenges: a fully manual, document-centric process with no simulation capability — where the only means to test a policy is to launch it in a limited area of KSA.

Expected Impact and Outcomes

- Reduce average policy development cycle from 6–12 months to under 6 months (target: 50% reduction by Year 2)
- Embed predictive simulation so policies requiring simulation under SDD-confirmed trigger rules are tested for fiscal, social, and market impact across multiple scenarios before enactment
- Eliminate policy silos through cross-sector conflict and ripple-effect detection across housing, municipal, fiscal, environmental, and smart-city domains
- Build institutional memory through a version-controlled Policy Registry with full assumption and decision traceability
- Systematically score policies against Vision 2030 KPIs, particularly the 70% homeownership target and fiscal sustainability goals

Success Metrics (Baseline → 6M / Y1 / Y2)

- Active platform users: 0 → 10 / 20 / 50
- Policies simulated through the platform: 0 → 3 / 5 / 10
- Average policy development time reduction: 0 → 10% / 30% / 50%
- Simulation accuracy (HITL approvals without edits): baseline 0; target 80% by Year 2 (per BRD Strategic and Operational Metrics)
- Conflict detection recalls prior to ministerial review: baseline 0; target $\geq 85\%$ by Year 2 (per BRD Strategic and Operational Metrics). Ground truth established through Maarifa/Sanad cross-referencing, MOMAH legal expert validation, and a labelled benchmark dataset
- Assumption documentation completeness: baseline 0; target 90% by Year 2 (per BRD Strategic and Operational Metrics). Metadata schema and required fields to be finalized in SDD.

- Automated risk threshold alerts: baseline 0; target >90% by Year 2 (per BRD Strategic and Operational Metrics). Fiscal/legal risk taxonomy and trigger rules to be finalized in SDD
- Policy simulation and conflict check coverage: baseline 0; target threshold to be confirmed in SDD after lifecycle trigger rules, simulation applicability, and conflict-check coverage requirements are finalized

Business Case

There are currently no automated processes or systems for policy launch at MOMAH. The Policy AI Lab replaces a fully manual process, targeting measurable efficiency gains: a 50% reduction in policy development cycle time by Year 2, reduction in post-approval policy reversals, and improved fiscal accountability through systematic impact simulation. A formal NPV/IRR financial model will be developed once simulation output formulas and fiscal assumptions are finalized with the client. Formula methodology: (1) NPV uses standard Discounted Cash Flow — $NPV = \sum (\text{net_cash_flow_t} / (1+r)^t)$, using MoF-published discount-rate inputs; the baseline analysis horizon shall follow the BRD fiscal-impact horizon, with any extended horizon to be confirmed with MOMAH/MoF in the SDD; (2) GDP contribution uses direct + indirect + induced economic impact methodology aligned with GASTAT national accounts and MoF input-output tables; (3) Social Equity Impact uses distributional analysis across GASTAT income deciles, MOMAH beneficiary data, and SAMA mortgage accessibility data. Financial input values (CAPEX, OPEX, IRR, NPV) to be provided by MOMAH once the simulation engine is calibrated.

AI Risk Classification: High Risk

The system directly informs policy decisions affecting national housing strategy, multi-sector regulatory alignment, and Vision 2030 national KPIs. All AI outputs are advisory only; human-in-the-loop validation is mandatory at every approval stage (Policy Analyst → Director → Minister). No policy recommendation reaches enactment without expert review and formal sign-off.

Key Assumptions Underpinning the Design

- MOMAH internal system data (housing permits, Sakani, Balady) will be made accessible to the platform
- Integration with Maarifa (legal references) and Sanad (legal advisory) systems is technically and contractually feasible within project scope
- MOMAH SSO integration is available and will be provided for authentication and role-based access control
- All AI inference and data storage will remain within KSA, in full compliance with PDPL and NCA Essential Cybersecurity Controls
- Human-in-the-loop governance framework will be formally adopted by MOMAH stakeholders as a mandatory part of the policy lifecycle

1. Introduction

1.1 Purpose

This High-Level Solution Design (HLSD) document provides a high-level blueprint for the Policy AI Lab solution at the Ministry of Municipalities and Housing (MOMAH), translating the business requirements documented in the BRD v2.0 into an initial solution design. It defines the solution purpose, scope, key capabilities, user personas, workflows and user journeys, business requirements, success metrics, and delivery roadmap. This document is intended to be reviewed and endorsed by the AI Solutions Steering Group before detailed design and development commences in Phase 2. It serves as the primary reference for all subsequent design, architecture, and delivery activities.

1.2 Scope

In Scope

The Policy AI Lab is delivered in three phased modules, each building on the previous, as defined in BRD §6.1:

- **Policy Registry & Drafting:** a unified, canonical Policy Registry with versioning and metadata (title, objectives, sectors, legal basis, KPIs, effective date); a Policy Project Concept Builder that validates alignment with sector and higher-level national strategies and auto-generates policy memoranda covering legal basis, justification and objectives, key provisions, financial/economic/social impact summaries and risk considerations, and proposes a recommended draft; amendments to existing policy with structured justification and traceability to source changes; and Institutional Memory storing policy iterations with full explainability of why specific levers were chosen.
- **Benchmarking & Conflict Detection:** policy benchmarking and comparative analysis against peer countries and global standards, contextualised for the Saudi economy, with suggested amendments and highlighted gaps and risks; policy conflict and duplication detection identifying overlapping, redundant or contradictory policies, flagging potential regulatory conflicts across sectors and providing impact mapping to affected policies; and cross-sector ripple-effect mapping of interdependencies across housing, municipal, fiscal and other economic sectors.
- **Policy Impact Simulation:** what-if scenario testing across economic, social and municipal variables, simulation of multiple policy scenarios, identification of associated risks and trade-offs, and sensitivity analysis based on configurable assumptions; consultation comment synthesis that ingests public and stakeholder feedback, clusters themes and recurring issues, summarizes key concerns and proposes structured amendment options; and a Risk Scoring Framework with a formal taxonomy (Legal, Fiscal, Operational, Strategic, International) and escalation thresholds.

Out of Scope

The following capabilities are explicitly excluded from the current phase, per BRD §6.2:

- **Autonomous Policy Approval:** the system will not have authority to approve, enact or publish policies; all final decisions remain human-led.
- **Binding Legal Interpretation:** the system does not provide legally binding interpretations or replace formal legal counsel; outputs are advisory and require expert validation.
- **Direct Legislative Draft Submission:** no automatic submission to legislative bodies or gazette systems without human validation and approval workflows.
- **Real-Time Macroeconomic Forecasting Infrastructure:** the system will not build or replace national econometric forecasting platforms; it relies on predefined models, assumptions or integrated external datasets.
- **Full Regulatory Enforcement Monitoring:** the system will not monitor real-time compliance or enforce regulatory actions; no inspection automation or enforcement execution.
- **Custom Model Training from Scratch:** the system will not develop foundational LLMs or large econometric models from scratch; it leverages existing AI models and configurable frameworks.
- **Political Decision-Making:** the system does not determine political priorities or make value-based policy choices; strategic direction remains with policymakers.
- **Operational Municipal Service Delivery:** the Lab will not manage day-to-day municipal operations (e.g., waste collection, utilities).
- **Citizen-Facing Applications:** the Lab is an internal decision-support tool, not a public-facing service.

1.3 Intended Audience

Audience	Purpose of Review
Central AI Committee	[For moonshot solutions only] Strategic review of an executive version of the HLSD and final stage-gate approval
AI Solutions Steering Group	Strategic and technical review of an executive version of the HLSD, stage-gate endorsement for moonshot solutions and stage-gate approval for standalone solutions
Sector Head	Review for alignment with sector needs and priorities, provide final approval before submission to Steering Group / Central AI Committee
AI Champion	Review for sector alignment
NHCI ARB	Review for technical architecture (no approval at this stage)

NHCI Technical Lead(s)	Day-to-day delivery management and review
Oversight team	Governance review, standards compliance, and delivery assurance across the program
AI Solution Owner	Solution-level owner; provide input, review key requirements, business outcomes, and user journeys
Solution Vendor	Document owner - produced and maintained by vendor. Responsible for completeness and accuracy of all sections.

Note: For governing bodies' review (i.e., Central AI Committee and AI Solutions Steering Group) an executive summary presentation needs to be developed covering all elements listed in the Executive Summary (at the beginning of this document)

1.4 Definitions & Acronyms

Term / Acronym	Definition
HLSD	High-Level Solution Design
NHCI	NHC Innovation
MOMAH	Ministry of Municipalities and Housing
ARB	Architecture Review Board
AI	Artificial Intelligence
Preparation phase	Define the vision, objectives, scope, requirements, and high-level solution approach
Prototype phase	Validate UX/UI, key features, and user journeys before full development
Alpha phase	Build and test core features in a sandbox environment to prove the technical approach
Beta phase	Complete feature development and validate the solution with integrations in a staging environment

Release & Stabilization phase	Deploy to production, release improvements, fix issues, and stabilize the solution
Operate phase	Hand over to operations/technical teams and drive adoption through change management and ongoing support
SDD	Solution Design Document (Phase 2 deliverable, follows HLSD approval)
BRD	Business Requirements Document
HITL	Human-in-the-Loop — governance model requiring mandatory human review of AI outputs before action is taken
ABM	Agent-Based Model — computational simulation model used to simulate the impact of housing and municipal policies
PDPL	Saudi Personal Data Protection Law — national data privacy regulation governing processing and storage of personal data within KSA
NCA	National Cybersecurity Authority — Saudi authority issuing Essential Cybersecurity Controls applicable to this solution
SDAIA	Saudi Data and AI Authority — national body providing AI ethics guidelines and governance framework
SIBOR	Saudi Interbank Offered Rate — key interest rate benchmark used as a simulation input variable
LTV	Loan-to-Value Ratio — mortgage lending parameter used as a policy lever in housing simulations
GASTAT	General Authority for Statistics — Saudi national statistics body providing demographic and economic data
SAMA	Saudi Arabian Monetary Authority — central bank providing financial and mortgage lending data
SSO	Single Sign-On — authentication mechanism enabling users to access the platform with their existing MOMAH credentials
DMS	Document Management System — MOMAH's existing system for storing, versioning, and archiving policy documents

XAI	Explainable AI — AI design principle ensuring model outputs include interpretable explanations accessible to non-technical users
KPI	Key Performance Indicator
Vision 2030	Saudi Arabia's national transformation plan, including a target of 70% homeownership — a primary alignment benchmark for the Policy AI Lab
HDOC	Housing Data and Operations Catalogue — MOMAH internal operational database confirmed as a required integration source for the Policy AI Lab. Contains MOMAH-specific operational records and housing programme data not available from external sources. Detailed database schema, integration interface specifications, and data quality assessment to be conducted in Phase 2 following a dedicated scoping session with MOMAH data owners.
EWG	Expert Working Group — MOMAH-led advisory body formed to provide and validate domain knowledge inputs for the Policy Simulation engine. Responsibilities include defining baseline policy assumptions, validating ABM parameters, and signing off on Low Confidence outputs. See full HLSD §3.7.2.
Policy Instrument	The specific mechanism — such as a tax, subsidy, regulation, land grant, fee, or eligibility threshold — used by a policy to achieve a stated objective. Referenced in full HLSD §3.3.6 (Simulation Input — Policy Variables) and full HLSD §3.3.5 (Benchmarking — Mechanism Type). Aligned with BRD Terms & Definitions.
Fiscal Neutrality	A state where the implementation costs of a policy are fully offset by the economic value, revenue uplift, or savings the policy generates. Used as a design constraint in CAP-02 Gap Analysis intervention recommendations (full HLSD §3.3.2) and in CAP-05 Benchmarking Fiscal Commitment scoring (full HLSD §3.3.5). Aligned with BRD Terms & Definitions.
Policy Knowledge Graph	A structured, machine-readable digital map of the relationships between policies, regulations, entities, sectors, Vision 2030 KPIs, and outcomes. Forms the backbone of the Policy Registry & Institutional Memory Layer (full HLSD § 5.2 Table 26 — Policy Registry & Knowledge Graph) and supports cross-sector ripple-effect detection by CAP-04 Conflict Detection Engine (full HLSD § 3.3.4). Aligned with BRD Terms & Definitions.

2. Business Context

2.1 Business Problem

MOMAH operates at the center of Saudi Arabia's national urban development, municipal governance, and housing strategy. Its mandate directly influences economic diversification, infrastructure efficiency, private sector participation, and quality of life — all foundational pillars of Vision 2030. As the Kingdom accelerates reform, the complexity, interdependency, and scale of policy decisions have significantly increased, yet MOMAH's current policy lifecycle is structurally unprepared for this level of systemic complexity.

Current State

Today, MOMAH's policy lifecycle is document-centric rather than system-centric, fragmented across data sources and stakeholders, dependent on manual analysis and siloed expertise, limited in predictive modeling capability, and reactive rather than simulation-driven. There are no automated processes for policy launch, and no system exists to support it. The only way MOMAH currently tests a policy is by launching it in a small area of KSA — a slow, costly, and high-risk approach to policy validation.

Five Core Pain Points

- **Limited System-Wide Visibility:** Policies are developed and reviewed without a unified, canonical registry. There is no single source of truth for active policies, their dependencies, or their status across ministries.
- **Insufficient Predictive and Simulation Capability:** Impact assessments are static reports. Scenario analysis is weak, particularly regarding risk identification and unintended consequences. Fiscal and social ripple effects are mostly either missed or not comprehensively analyzed.
- **Weak Institutional Memory and Feedback Loops:** Lessons from past policies are not systematically captured. There is no structured mechanism to trace assumptions and outcomes or compare policy performance over time.
- **Limited Global Benchmarking Integration:** International standards and peer-country benchmarks (e.g., Singapore, Norway) are not embedded into policy design in a structured way, limiting alignment with global best practices.
- **Extended Policy Development Timelines:** The current process requires many iterations and significant stakeholder bandwidth, resulting in policy development cycles of 6–12 months.

Policies now affect multiple sectors simultaneously — real estate, financial, industrial/construction, environmental, social, and smart-city sectors — carrying long-term fiscal and macroeconomic implications. Without a system capable of modeling these interdependencies, MOMAH risks regulatory conflicts, fiscal overruns, and failure to achieve Vision 2030 targets, including the 70% homeownership goal.

2.2 Strategic Alignment

Vision 2030	Enhance government effectiveness through data-driven policymaking, improved regulatory coherence, and accelerated reform delivery —
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Objective	directly supporting the 70% homeownership target, Quality of Life improvement, and fiscal sustainability under Vision 2030.
NHCI Strategic Objective	Advance AI-powered decision-intelligence capabilities within MOMAH, operationalizing AI for governance, institutional learning, and predictive policy analysis in support of NHCI's AI programme goals.
MOMAH Sector Strategic Objectives (SSOs)	Strengthen evidence-based policy development, improve regulatory quality, reduce policy contradictions and sectoral fragmentation, accelerate policy development cycles, enhance housing affordability and supply planning through advanced simulation, and build institutional memory for long-term governance excellence.
MOMAH AI Strategy Alignment	Deploy AI-driven decision-support tools that enhance policy quality, predictive capability, and institutional efficiency. Operationalize AI for governance, insight generation, and institutional learning through automated drafting, conflict detection, multi-variable simulation, risk scoring, and continuous model improvement — all under a human-in-the-loop governance framework compliant with SDAIA AI Ethics Guidelines.

2.3 Key Stakeholders

Stakeholder	Role	Interests / Concerns	Responsibility	Name	Contact
AI Solution Owner	Solution-level owner	Requirements definition; ensure progress in line with business needs	Participate in vendor check-ins, Sprint Reviews, Steering Group, and design discussions	Abdulmalik Alshabana	a.shabanat@mamah.gov.sa
AI Champion	Sector-level coordinator	Cross-solution coordination; surface risks, dependencies, and adoption needs within sector	Participate in Sprint Reviews and Bi-Weekly Sector meetings	Hamad Alsharida	halsharida@mamah.gov.sa

Sector Head	Sector-level sponsor	Solution direction, progress alignment, escalation support	Participate in Bi-Weekly Sector meetings and Central AI Committee sessions	Majid bin Abdullah Al-Hogail	
Policy KPI Owner / Business Owner	MOMAH-side owner of KPI targets and policy outcomes; primary point of contact for business requirements and metric definition	Simulation accuracy, conflict detection quality, assumption completeness, risk alert coverage, and overall policy outcome measurement	Own and validate all KPI targets; approve simulation methodology and metric definitions; participate in Sprint Reviews and milestone sign-offs	Abdulmalik Alshabana t	a.shabanat@momah.gov.sa

2.4 Expected Business Outcomes

- **Policy Quality Improvement:** Fewer contradictions and higher evidence standards across all MOMAH policies; systematic Vision 2030 alignment scoring for every policy draft.
- **Risk Reduction:** Early identification of unintended consequences, fewer regulatory escalations post-launch, and automated fiscal and legal risk alerts for policies exceeding defined thresholds.
- **Accelerated Reform Cycles:** Reduction in average policy development time from 6–12 months to under 6 months, enabling MOMAH to respond faster to market volatility, regulatory demands, and Vision 2030 milestones.

2.5 Current Solution Landscape

Solution Name	Owner / Provider	Dedicated Budget	Time-line / Status	Core features	Coverage of Problem (2.1)	Gaps / Limitations	Demo / Documentation
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None (No existing system)	N/A	N/A	N/A — no system currently in use	N/A — there are no automated tools, AI systems, or digital platforms currently in use for policy drafting, simulation, conflict detection, or benchmarking.	None — the business problem is entirely unaddressed by any existing digital system.	All policy development activities are conducted manually using documents and spreadsheets. Simulation is performed by launching a policy in a small geographic area of KSA — a slow, costly, and irreversible validation method. There is no registry, no conflict detection, no institutional memory, and no AI assistance of any kind.	N/A
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Initiative	Timeline	Owner / Provider	Approach taken	Outcome	Key Reason for Failure	Lessons Learned for Current Design	Demo / Documentation
N/A — no previous digital or AI initiative has been attempted for policy management at MOMAH.	N/A	N/A	N/A	N/A	N/A	N/A	N/A

3. Solution Overview

3.1 Solution Description

The Policy AI Lab is a centralized, AI-powered decision-intelligence platform that transforms MOMAH's policy lifecycle from fragmented and reactive to unified, predictive, and evidence-based. It provides government users — from Policy Analysts to the Minister — with role-specific tools to draft, simulate, conflict-check, benchmark, and approve policies within a single governed environment.

The platform operates across three integrated capability layers: (1) a canonical Policy Registry and AI-assisted drafting environment, enabling structured policy creation with legal basis extraction, Vision 2030 alignment validation, and assumption documentation; (2) a Benchmarking and Conflict Detection engine, automatically identifying overlapping or contradictory policies across sectors and benchmarking against peer countries such as Singapore and Norway; and (3) a Multi-Scenario Simulation engine, enabling Housing Economists to stress-test policies across economic, demographic, supply, and market variables — including interest rate shocks, land price changes, and developer behavior — before any policy reaches ministerial review.

All AI outputs are advisory. A tiered human-in-the-loop governance framework (Policy Analyst → Director → Minister) ensures no AI recommendation reaches enactment without expert validation. The platform is designed for full compliance with Saudi PDPL, NCA Essential Cybersecurity Controls, and SDAIA AI Ethics Guidelines, with all data and AI inference remaining within KSA.

3.2 User Personas

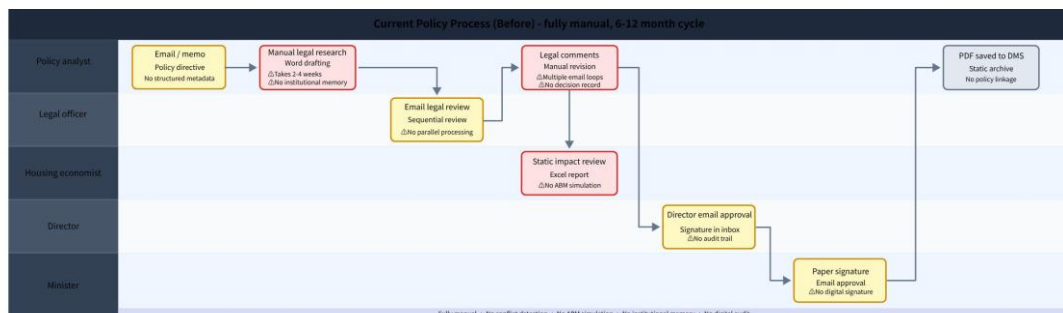
Persona	Description	Primary Need	Interaction with the Solution	Frequency of Use
Minister (H.E.)	CEO of MOMAH; ultimate decision-maker for all ministry policies.	High-level dashboard of policy health, Vision 2030 alignment scores, risk alerts, and Tier-3 escalations requiring ministerial decision.	Executive Intelligence Cockpit: scenario comparison, final approval gateway, override capability.	Weekly / On escalation
Senior Leader / Executive	High-level decision-makers within MOMAH's Executive Office; responsible for strategic direction and cross-ministerial alignment.	Portfolio view of all active policies, cross-sector ripple effects, regional impact visualization, and Tier-2 strategic briefings.	Policy portfolio dashboard, Vision 2030 Alignment Auditor, Regional Impact Heatmaps.	Weekly

Policy Analyst	Primary strategist responsible for drafting and refining policy frameworks.	AI-assisted policy drafting with legal basis extraction, gap analysis, consultation comment synthesis, and access to institutional memory.	Policy drafting sandbox, version control, assumption builder, knowledge registry.	Daily
Housing Economist	Expert focused on the financial and macroeconomic impact of policy levers.	Multi-variable simulation lab, Black Swan stress testing, sensitivity analysis, and ability to manipulate macro, demographic, supply, and market variables.	Agent-Based Simulation engine, scenario configurator, risk escalation monitoring.	Daily / Per policy
Data Scientist / AI Unit	Technical team responsible for the platform's underlying models, data pipelines, and AI governance.	Model retraining, explainability audits, drift detection, raw log access, threshold configuration, and model versioning.	Model governance dashboard, retraining pipeline, explainability audit tools.	Weekly / On drift alert
Legal & Compliance Officer	Gatekeeper ensuring all policy drafts adhere to Saudi Law, Royal Decrees, and PDPL/NCA requirements.	Regulatory conflict detection, legal basis extraction, clause-level traceability, and compliance gateway.	Regulatory Conflict Engine, Maarifa/Sanad integration interface, override permissions.	Per policy / On conflict alert
Business Owner / Product Manager	Responsible for strategic direction and delivery of the Policy AI Lab	Policy lifecycle orchestration, workflow management, system usage	System health dashboard, workflow configurator, usage analytics.	Daily

	solution.	analytics, KPI and threshold configuration, and role-based access control.		
Stakeholder (Internal/External)	Representatives from other ministries or the public providing consultation feedback.	Consultation portal to submit feedback, view summarized responses, and track how input influenced policy drafts.	Feedback submission interface, consultation synthesis viewer.	Per consultation cycle
Simulation Expert	A technically qualified expert invoked for complex or high-stakes policy simulations that exceed the standard Housing Economist workflow.	Deep access to ABM model parameters, Monte Carlo configuration, calibration datasets, and scenario sensitivity controls; ability to document and justify parameter adjustments.	Advanced Simulation Lab access; ability to override default parameter bounds with written justification; responsible for confirming technical validity of simulation outputs before they are presented to the Policy Analyst for HITL review.	On demand — engaged only when simulation complexity, policy risk level, or Monte Carlo instability exceeds standard Housing Economist scope.

3.3 Process / Workflow

Current Policy Development Process



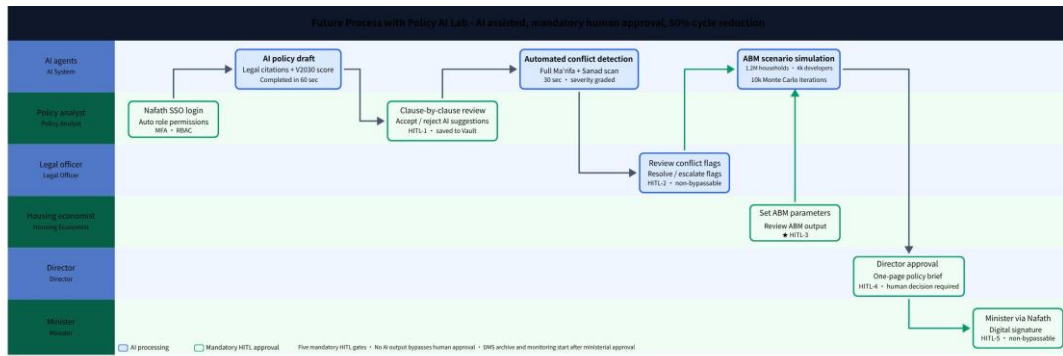
Current State — Policy Development Without the Policy AI Lab (Before)

MOMAH's current policy lifecycle is entirely manual and document centric. There are no automated systems, registries, or simulation tools in use. The process follows these steps:

1. **Policy Initiation (Manual):** A Policy Analyst receives a directive from a director or the Minister's Office, typically via email or physical memo. Policy context, relevant legislation, and prior precedents are located manually by the Analyst through personal knowledge, hard-copy archives, or informal queries to colleagues.
2. **Research and Drafting (Manual):** The Analyst research comparable policies and legal references by browsing government websites, reviewing physical law books, or consulting with the Legal Department directly. Policy drafts are written in Microsoft Word from scratch, with no structured template or institutional memory to draw from. This stage typically takes 2–4 weeks per policy.
3. **Internal Review (Manual, Sequential):** The Word document is circulated via email to Legal Officers, Department Heads, and other relevant departments for review. Comments are collected via email reply to threads or tracked-change annotations on Word files. Multiple review cycles are sequential (one reviewer at a time), with no parallel processing. Conflicts between reviewers are resolved in ad-hoc meetings with no formal documentation of decisions.
4. **Stakeholder Consultation (Manual, Ad Hoc):** Where required, public or inter-ministerial consultation is conducted via email distribution, printed questionnaires, or in-person workshops. Feedback is manually collated and summarized with no structured synthesis tool. This stage can take 4–8 weeks.
5. **Impact Assessment (Manual, Limited):** Fiscal and social impact assessments are produced as static Word or Excel reports by internal economists or external consultants. Scenario analysis is limited; the primary method for testing a policy's real-world impact is to launch it in a small geographic pilot area — a slow, costly, and effectively irreversible validation method.
6. **Approval Workflow (Manual, Paper/Email):** The drafted policy is submitted for Director approval via email, then escalated to the Minister's Office. Sign-off is given via email confirmation or physical signature. There is no formal digital audit trail; approval records are maintained in individual email inboxes or physical files.
7. **Policy Storage and Archiving (Fragmented):** Approved policies are stored in a Document Management System (DMS) as static PDF/Word files with limited metadata. There is no version control, no cross-policy linkage, and no structured mechanism to identify conflicts with existing policies. Institutional memory resides in individuals, not systems.

Key pain points of the current state: No simulation capability; policy conflict detection is entirely manual and dependent on individual expertise; no institutional memory or cross-policy knowledge graph; policy development cycles of 6–12 months; no vision 2030 alignment scoring; no structured audit trail of AI or data-informed decisions (as no AI tools are currently used).

Future Policy Development Process



Future State — Policy Development with the Policy AI Lab (After)

The Policy AI Lab transforms the policy lifecycle from a sequential, document-centric process into a governed, AI-augmented, parallel workflow with mandatory human oversight at every critical decision point. Key changes are highlighted below, mapped to the capabilities in full HLSD §3.3 and the user personas in Section 3.2.

- Policy Initiation (CAP-01, CAP-02 — Policy Analyst):** Analyst authenticates via Nafath SSO; lands on a role-specific dashboard showing active policies, notifications, and action items. Opens the Policy Registry and creates a new policy record with structured metadata: name, target sector, Vision 2030 pillar(s), budget envelope, and legal constraints. The system validates uniqueness against the existing Policy Registry and checks for naming conflicts inline. *Change from current state: Structured initiation with metadata enforcement replaces informal email directive; institutional memory immediately available.*
- AI-Assisted Drafting (CAP-03 — Policy Analyst + LLM):** Analyst selects a grounding mode (Precedent-heavy / Balanced / Novel) and clicks Generate Draft. The Policy Drafting Agent queries the Knowledge Graph for relevant Royal Decrees, historical precedents, and Vision 2030 KPI linkages, then generates a structured draft within 60 seconds. The draft includes a Vision 2030 Alignment Score (0–100), an Initial Risk Score, and a clause-by-clause Explainability Report citing all data sources and legal references. Analyst reviews each clause, accepts or rejects AI suggestions, and saves the version to the Institutional Memory Vault. *Change: Days-to-weeks of manual research compressed to 60-second AI-grounded draft generation; full explainability and source attribution mandatory.*
- Parallel Conflict Detection (CAP-01, CAP-04 — Conflict Detection Agent + Legal Officer):** Upon saving, the Conflict Detection Agent automatically scans the draft against the full Policy Registry and KSA legal corpus (Maarifa, Sanad) in the background. A Conflict Report is generated within 30 seconds, categorized by severity (Critical/High/Medium/Low). Legal Officer reviews and resolves each flag: selecting Resolved, Requires Amendment, or Dismissed (Critical dismissals require written justification). Cross-ministry conflicts generate a cross-sector Ripple Effect report and notify affected ministries via Yesser GSB. *Change: Automated parallel conflict detection replaces sequential manual legal review; 30-second scan vs. weeks of manual analysis.*
- Multi-Scenario Simulation (CAP-06 — Housing Economist + ABM):** Once all Critical conflicts are resolved, the workflow evaluates whether the policy meets the SDD-confirmed simulation trigger conditions defined in full HLSD §6.4.1. Where a trigger applies, the Housing Economist opens the Simulation Lab and configures the variable matrix: Macro-Economic

(SIBOR, CPI), Demographic (household formation, urbanization), Supply (land tax, developer incentives), and Market (LTV ratio, rental yield) variables. Scenario definitions may include Baseline, Optimistic, Pessimistic, and Black Swan, subject to the policy type and approved simulation scope. The ABM engine runs scenario simulations using MOMAH-confirmed agent categories (households/residents, developers, banks and financial institutions, government/regulators, municipalities/regional authorities, investors, and market-environment variables). Scale, iteration counts, and Level-2/Level-3 runtime targets are subject to SDD data assessment, SDAIA platform benchmarking, and MOMAH approval. Level-1 is treated as a quick-estimate or simplified standard scenario aligned to the BRD 60-second target. Outputs include Predictive Success Score, Homeownership impact, Price-to-Income ratio, BRD-aligned fiscal NPV, and a regional heatmap where applicable. If Monte Carlo failure rate exceeds 30%, the policy is labelled Unstable and blocked. Policies that do not meet simulation-trigger criteria must still complete required conflict, legal, and governance checks. *Change: Replaces the current approach of piloting a policy in a small geographic area; provides simulation-based impact testing where the agreed trigger rules require it before real-world enactment.*

5. Tiered HITL Approval Workflow (CAP-09 — Policy Analyst, Director, Minister):

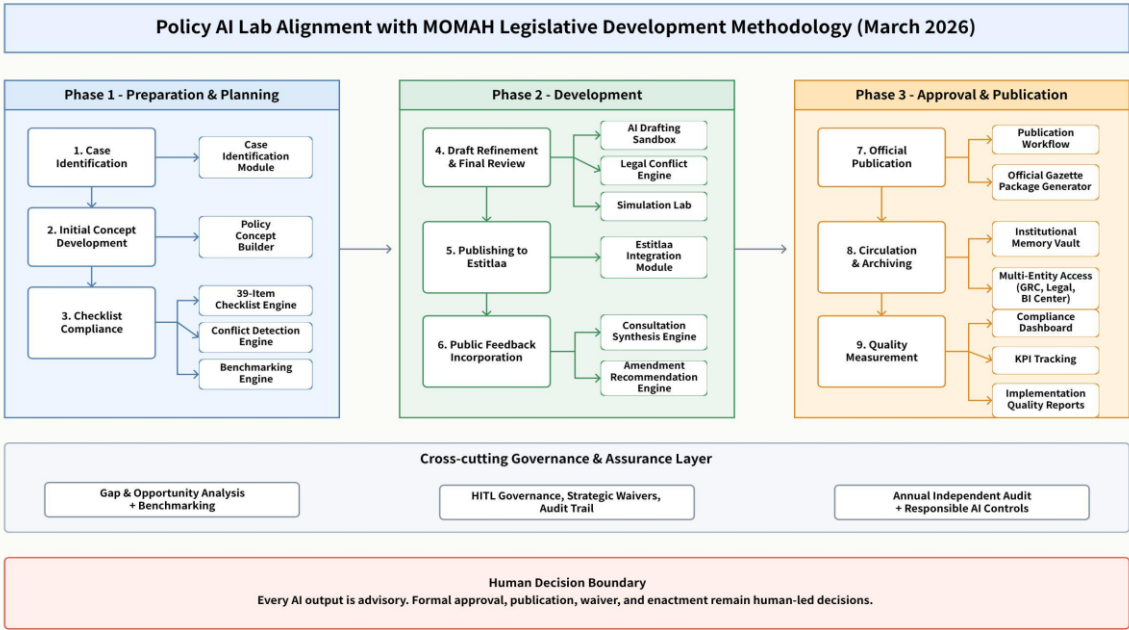
Required lifecycle outputs route to the HITL Review stage, including simulation outputs where simulation trigger conditions apply. The Policy Analyst reviews and submits for Director approval (Tier 2 briefing: one-page policy summary with risk flags and scenario comparison). Director reviews and approves or requests revision. On Director approval, the policy routes to the Ministerial Executive Intelligence Cockpit for final decision. The Minister reviews the Decision Summary, Regional Impact Heatmap, and scenario comparison; selects Approve, Reject, Request Revision, or Request Further Simulation; and digitally signs via Nafath. All decisions are logged immutably. *Change: Structured, documented, multi-stage approval with digital signatures and immutable audit trail replaces email-based sign-off.*

6. Approval Archiving and KPI Tracking (CAP-10, CAP-11): On Ministerial approval, policy status changes to Enacted; DMS automatically archives all policy documents, impact briefs, and decision records. The Risk Escalation Agent tracks configured KPI/outcome indicators against the simulated baseline where approved data feeds are available; threshold breaches trigger tiered review alerts (Analyst, Director, or Minister) to support human-led policy assessment. It does not perform enforcement monitoring or automated regulatory action. *Change: Post-launch monitoring replaces reactive response to policy failures; institutional memory preserved for future policy design.*

Summary of workflow transformation:

Dimension	Before (Current State)	After (Policy AI Lab)
Policy draft generation time	2–4 weeks (manual research)	60 seconds (AI-grounded draft)
Conflict detection	Manual, weeks, dependent on individual expertise	Automated, 30 seconds, full legal corpus coverage
Impact simulation	Geographic pilot only (slow, costly, irreversible)	Trigger-based multi-scenario ABM simulation across 13

		regions where required before enactment
Policy development cycle	6–12 months	Target: under 6 months (50% reduction by Year 2)
Institutional memory	Fragmented in individual email inboxes and physical files	Centralized Institutional Memory Vault with full decision traceability
Approval audit trail	Email threads and physical signatures	Immutable digital audit log with Nafath-signed decisions
Vision 2030 alignment scoring	None (informal judgement)	Automated 0–100 score per policy draft, clause-level breakdown



3.4 User Journey

ID	User Journey	Summary	Steps
J-01	Policy Analyst: Draft a New Policy	A Policy Analyst initiates a new housing policy. The AI generates a structured draft grounded in Saudi Law and Vision 2030 KPIs. The Analyst reviews	1. Analyst authenticates via Nafath SSO; role-based landing page presented. 2. Opens Policy Registry → 'Create New Policy'. 3. Completes required fields: Policy Name, Target

		clause-by-clause and saves the draft for conflict review.	Sector, Sub-Sector, Primary Objective, Vision 2030 Pillar(s), Budget Envelope, optional constraints. 4. Selects Grounding Mode and clicks 'Generate Draft'. 5. AI Policy Drafting Agent queries Knowledge Graph for Royal Decrees, precedents, and KPIs (≤60 sec). 6. System presents Draft Preview with Explainability Report, Vision 2030 Alignment Score (0–100), and Initial Risk Score. 7. Analyst reviews and edits individual clauses; accepts or rejects AI suggestions. 8. Clicks 'Save to Vault' — stored as v0.1; status set to 'Draft'. System autoroutes to Conflict Detection. AI: Steps 5–6 (LLM + KG retrieval). HITL: Steps 7–8 mandatory.
J-02	Housing Economist: Run a Multi-Scenario Policy Simulation	Where SDD-confirmed simulation trigger conditions apply, a Housing Economist runs ABM simulations on an approved draft to forecast fiscal, housing supply, and affordability impacts across multiple scenarios before ministerial review.	1. Economist opens Simulation Lab; selects a policy draft with status 'Simulation Required' or 'Ready for Simulation'. 2. Configures Variable Matrix: Macro-Economic (SIBOR delta, CPI), Demographic (household formation, urbanization), Supply (land tax, developer incentives), Market (LTV ratio, rental yield). 3. Defines Scenario Definitions: Baseline, Optimistic, Pessimistic, and Black Swan (e.g., SIBOR +300bps + material cost +40%). 4. Clicks 'Run Simulation'. ABM + Monte Carlo executed according to the simulation level selected; Level-1 is treated as a quick-estimate/simplified standard scenario targeted at ≤60 seconds, with detailed Level-

			2/Level-3 performance to be benchmarked in SDD. 5. System displays, where applicable: Predictive Success Score (0–100), Confidence Band, Homeownership impact, Price-to-Income ratio, Fiscal NPV (15-year baseline horizon per BRD; extended scenarios subject to MOMAH/MoF confirmation), Supply Velocity, and regional heatmap. 6. If Monte Carlo failure rate >30% → policy labelled 'Unstable', blocked from Executive Dashboard. If NPV negative → MoF Strategic Waiver required. 7. Economist reviews outputs via drill-down, exports Impact Brief PDF. 8. If Success Score ≥80% → policy advances to HITL Review. If <80% → Failure Explanation generated; Economist may re-simulate. AI: Steps 4–5 (ABM + Monte Carlo, where simulation is triggered). HITL: Steps 6–8 mandatory for simulation-triggered policies.
J-03	Legal Officer: Review Conflict Detection Results	A Legal & Compliance Officer reviews AI-generated conflict flags, resolves or escalates each flag, and clears the policy for simulation-trigger evaluation or the next applicable lifecycle stage.	1. Officer receives notification: 'Conflict Detection Complete — Policy X requires review.' 2. Opens Regulatory Conflict Engine; Conflict Report displayed by severity (Critical/High/Medium/Low) with source articles and affected ministries. 3. For each flag: reviews conflicting clause, source article/Royal Decree reference, severity, and AI recommendation. 4. Selects resolution: Resolved / Requires Amendment / Dismissed (Critical dismissals require

			written justification). 5. Cross-ministry conflicts generate Cross-Sector Ripple Effect report; affected ministry notified via Yesser GSB. 6. Contested flags escalated to Director of Policy (Tier 2 briefing). 7. All Critical flags resolved/escalated → Officer clicks 'Clear Conflict Review'; workflow evaluates SDD-confirmed simulation trigger rules and routes the policy to Simulation Required, HITL Review, or the next applicable lifecycle stage; all resolutions logged immutably. AI: Steps 2–3 (automated conflict scan; Conflict Report in ≤30 sec). HITL: Steps 4–7: every flag requires explicit human decision.
J-04	Minister: Review and Approve a Policy	The Minister reviews an AI-analyzed policy package in the Executive Intelligence Cockpit, including simulation outputs where applicable, and issues a final digitally-signed approval decision.	1. Minister opens Executive Intelligence Cockpit via Nafath SSO. Landing screen: Policy Health gauge, Vision 2030 trajectory chart, policy tiles awaiting decision. 2. Selects a policy tile; one-page Decision Summary shown: Alignment Score, top-3 risks, fiscal NPV where applicable, and scenario comparison where simulation was triggered. 3. Reviews Regional Impact Heatmap where simulation or regional impact analysis is available. 4. If resilience <50% or NPV negative, system presents Recovery Scenarios for comparison where simulation scope supports them. 5. Active Tier-3 critical alerts from MOMAH-approved KPI and policy-risk data feeds displayed

			prominently (e.g., '22% drop in developer participation under current incentive structure'). 6. Minister selects: Approve / Reject / Request Revision / Request Further Analysis or Simulation. 7. Decision digitally signed via Nafath; logged immutably with rationale. 8. On Approval: status 'Enacted'; DMS archives all documents; post-launch KPI/outcome tracking begins. On Rejection: policy returned to Analyst with comments. AI: Steps 2–5 (alignment score, risk intelligence, and simulation outputs where applicable). HITL: Steps 6–7: final decision is mandatory human action.
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ID	User Journey Step (J-01: Policy Analyst — Draft a New Policy)	System / AI Response
J-01.1	Analyst authenticates via Nafath SSO	System validates identity via Nafath MFA; retrieves role claims from MOMAH AD; presents Policy Analyst landing page with active drafts, notifications, and quick-access tools
J-01.2	Analyst opens Policy Registry and clicks 'Create New Policy'	System presents the Policy Drafting Workspace with required and optional fields; schema validation rules pre-loaded
J-01.3	Analyst completes Policy Name, Target Sector, Sub-Sector, Primary Objective, Vision 2030 Pillar(s), Budget Envelope (mandatory); Legal/Geographic Constraints (optional)	System validates uniqueness of Policy Name in registry; checks for duplicate names inline; prevents save if mandatory fields are empty
J-01.4	Analyst selects Grounding Mode (Precedent-heavy / Balanced / Novel)	System triggers Policy Drafting Agent; spinner + ETA shown. Agent queries

	and clicks 'Generate Draft'	Knowledge Graph for relevant Royal Decrees, historical precedents, Vision 2030 KPI linkages, and sector benchmarks. LLM generates structured draft
J-01.5	Analyst reviews the generated Structured Draft Preview	System presents: (1) Draft with clause-by-clause Explainability Report (expandable per clause, citing data sources and legal references); (2) Vision 2030 Alignment Score (0–100 gauge, color-coded); (3) Initial Risk Score; (4) 'Insufficient Grounding' flag if the citation minimum (≥ 3 historical precedents or ≥ 2 legal circulars per BRD Explainability Gateway, full HLSD §6.3) is not met
J-01.6	Analyst reviews each clause, edits where needed, accepts or rejects AI suggestions	System updates draft in real-time; tracks analyst edits; version-diff shown against AI-generated original
J-01.7	Analyst clicks 'Save to Vault'	System persists draft as v0.1 in Institutional Memory Vault; policy status set to 'Draft'; confirmation toast shown. System immediately auto-routes draft to Conflict Detection Agent for background scanning
J-01.8	Analyst receives Conflict Detection notification (async)	System notifies Analyst when conflict scan is complete; conflict count shown by severity. Policy status updated to 'Conflict Review Required', 'Simulation Trigger Evaluation', 'Simulation Required', or the next applicable lifecycle stage depending on results and SDD-confirmed trigger rules

4. Requirements Summary

This section summarizes the key business and technical requirements. The complete requirement set is maintained in DOC-01 Product Requirements (see Additional resources).

4.0 BRD–HLSD Capability Traceability

The table below maps the four high-level capabilities defined in the BRD to the thirteen mapped HLSD capabilities (CAP-01 to CAP-13) listed in the traceability tables below (detailed capability design in the full HLSD §3.3). This traceability confirms full coverage of BRD scope while distinguishing mapped capability coverage from the detailed design subsections that follow.

BRD Capability	BRD CAP	HLSD Capabilities	HLSD CAP(s)
Policy Intelligence & Knowledge Layer	BRD CAP-01	Policy Registry; Gap & Opportunity Analysis Engine	CAP-01, CAP-02
AI-Assisted Policy Generation	BRD CAP-02	Policy Concept Builder & AI Drafting Sandbox; Conflict Detection; Benchmarking & Comparative Policy Analysis Engine	CAP-03, CAP-04, CAP-05
Policy Simulation & Scenario Modeling	BRD CAP-03	Policy Simulation & Scenario Modeling	CAP-06
Collaboration & Workflow Orchestration	BRD CAP-04	Consultation Synthesis & Amendment Recommendation; Estitlaa Consultation Publishing Support; HITL Governance & Strategic Waiver; Approval Tracking & Formal Archiving; Post-Implementation Quality Measurement; Executive Decision Intelligence; Responsible AI & Auditability	CAP-07, CAP-08, CAP-09, CAP-10, CAP-11, CAP-12, CAP-13

In addition to the BRD mapping above, the table below traces the fourteen named agents in the approved AI-Strategy implementation card (two families: Project Study Agents and Drafting & Generation Agents) to the HLSD capabilities, confirming coverage and surfacing gaps for Stage-Gate alignment.

Family	Implementation-Card Agent	HLSD Capability (CAP)	Coverage status
Project Study	Policy Update Flagging Agent	CAP-04 (conflict scan + proactive monitoring), CAP-02	Covered — reactive from Alpha; proactive monitoring & trend prediction phased to Beta (BR-19)
Project Study	Project Concept	CAP-03	Covered

	Builder Agent		
Project Study	Strategic Alignment Agent	CAP-02 + full HLSD §3.8.2 Vision 2030 scoring	Covered
Project Study	Benchmark Comparator	CAP-05	Covered
Project Study	Operational Impact Analyzer	CAP-06 (operational/regulatory impact)	Covered — CAP-06 extended with operational/regulatory impact analysis (BR-20)
Project Study	Socioeconomic & Behavioral Impact Simulator	CAP-06	Covered
Project Study	Sensitivity & What-If Explorer	CAP-06	Covered
Project Study	Implementation Risk & Readiness Evaluator	CAP-06 (implementation-readiness dimension)	Covered — CAP-06 risk-scoring extended with implementation-readiness dimension; institutional inputs from MOMAH, delivered Beta (BR-22)
Drafting & Generation	Automated Policy Generator Agent	CAP-03	Covered
Drafting & Generation	Conflict & Duplication Detector Agent	CAP-04	Covered
Drafting & Generation	Policy Dependency Locator Agent	CAP-04 / CAP-01 (MOMAH + cross-entity)	Covered — MOMAH corpus from Alpha; cross-entity phased to Beta, subject to inter-ministerial data access (BR-21)
Drafting & Generation	Explanatory Memorandum Generator	CAP-03 / full HLSD §3.8.1	Covered

Drafting & Generation	Decision Package Assembler Agent	CAP-09 / CAP-10	Covered
Drafting & Generation	Consultation Comments Synthesizer Agent	CAP-07 / CAP-08	Covered

All fourteen agents are now dispositioned to a mapped capability and business requirement (BR-19–BR-22 in §4.1). Proactive monitoring, cross-entity dependency, and the implementation-readiness dimension are phased to Beta as noted in the coverage status above.

4.1 Business Requirements

ID	Requirement	Priority	Source
BR-01	Policy Registry & Centralized Document Management — the system shall provide a centralized repository for all MOMAH policies with version control, metadata tagging, full-text search, legal-basis linkage, decision history, final approved texts, and supporting records.	P0	BRD §3
BR-02	AI-Assisted Policy Drafting — the system shall generate a structured initial policy draft within 60 seconds based on policy officer inputs, using a domain-specific LLM and bilingual templates (Arabic / English). Draft outputs shall include legal basis, objectives, financial/social/economic impact summaries, risk considerations, and source-grounded explainability.	P0	BRD §4
BR-03	Human-in-the-Loop (HITL) Review Workflow — all AI-generated drafts, checklist results, simulation outputs, consultation amendment recommendations, waiver requests, and archival records shall pass through mandatory authorized human review and approval; no policy may be published without formal human sign-off.	P0	BRD §5; HITL governance controls
BR-04	Role-Based Access Control (RBAC) & Audit Trail — the system shall enforce	P0	BRD §6 / MOMAH IT / PDPL

	role-based permissions and maintain tamper-proof audit logs for all user actions, AI outputs, model runs, checklist validations, consultation decisions, waiver approvals, archival actions, and overrides.		
BR-05	Policy Impact Simulation (ABM) — the system shall simulate projected outcomes of draft policies using Agent-Based Modeling across household, demographic, and economic agent populations derived from KSA national datasets (GASTAT, SAMA, MoF).	P0	BRD §4 / KPI Owners
BR-06	Policy Benchmarking & Comparative Analysis — the system shall enable structured comparison of MOMAH draft policies against GCC, OECD, and international peer counterparts, with data freshness indicators, normalization assumptions, output gap tables, risk flags, and best-practice recommendations.	P0	BRD §4
BR-07	Conflict Detection & Compliance Checking — the system shall automatically identify regulatory conflicts, duplications, contradictions, and checklist gaps between draft policies, existing MOMAH policy records, KSA legal corpus, Maarifa, Sanad.	P0	BRD §4
BR-08	Arabic-English Bilingual Support — all user-facing interfaces, policy drafts, consultation synthesis, reports, dashboards, and generated packages shall support Arabic (RTL) and English; the AI drafting and NLP engines shall operate in both languages.	P1	BRD §3 / All Users
BR-09	Enterprise System Integration — the system shall integrate with MOMAH HCM and ERP for user identity and workflow routing; SSO via MOMAH identity	P2	BRD §6 / MOMAH IT

	provider (SAML 2.0 / OIDC) is required.		
BR-10	Case Identification & Policy Gap Analysis — the system shall support identification of cases requiring new or updated legislation and detect regulatory white spaces by comparing Policy Registry coverage against Vision 2030 KPIs, sector priorities, and peer-country benchmarks.	P0	BRD UC-16
BR-11	Policy Project Concept Builder — the system shall capture policy trigger, target sector, legal basis, strategic alignment, stakeholders, expected impacts, and constraints, then generate an initial concept note and policy memorandum for analyst review.	P0	BRD concept-builder requirements
BR-12	Consultation Comment Synthesis — the system shall ingest consultation feedback from configured channels, including Estitlaa where available, and provide NLP topic clustering, sentiment scoring, representative comments, structured amendment options, and traceability from comment themes to draft changes.	P0	BRD consultation requirements
BR-13	Estitlaa Consultation Publishing Support — the system shall prepare and validate draft consultation packages, metadata, consultation windows, deadlines, and handoff/status tracking for Estitlaa. Actual submission, publication, and enactment remain outside automated decisioning and require official MOMAH workflow execution.	P1	BRD consultation requirements
BR-14	Approval Tracking & Formal Archiving — the system shall capture and track the ministerial approval decision and its outcome for each policy, and archive the final approved policy text, decision record, and supporting material — with status and timestamps — in the Institutional	P1	BRD §3

	Memory Vault. Enactment and publication remain human-led and outside automated decisioning.		
BR-15	Post-Implementation Quality Measurement — the system shall track KPI/outcome performance, implementation quality, actual versus simulated outcomes, variance indicators, and periodic reports for GRC, Senior Leaders, and the Minister. It shall not perform regulatory enforcement monitoring or execute compliance actions.	P1	BRD UC-17
BR-16	Strategic Waiver Workflow — the system shall support fiscal waiver and strategic exception requests, MoF alignment where required, approval routing, decision rationale capture, and audit trail.	P1	BRD waiver references
BR-17	Executive Decision Intelligence — the Minister and Senior Leader dashboards shall include approved alert notifications, regional impact views, and configured policy-risk intelligence based on MOMAH-approved data feeds. The platform shall not provide real-time macroeconomic forecasting infrastructure.	P2	BRD executive dashboard references
BR-18	Annual Independent Audit — the system and governance model shall support annual independent audit of policy quality, benchmarking accuracy, model performance, and compliance with high-risk AI governance controls.	P1	BRD audit requirement
BR-19	Proactive Policy-Impact Monitoring — the system shall monitor newly issued laws, decrees and regulatory decisions, flag impacted existing MOMAH policies with justified amendment proposals and predict potential policy-impact areas via trend analysis. Reactive impact detection from Alpha; proactive monitoring and	P1	AI-Strategy Implementation Card (Policy Update Flagging Agent)

	trend prediction delivered in Beta.		
BR-20	Operational & Regulatory Impact Analysis — the system shall assess a draft policy's effect on operational processes and existing regulations and identify required compliance adjustments and efficiency improvements.	P1	AI-Strategy (Operational Impact Analyzer)
BR-21	Cross-Entity Policy Dependency — the system shall identify existing MOMAH policies impacted by a draft (with justification and suggested amendments) and, where inter-ministerial data access permits, other entities' policies. Cross-entity coverage phased to Beta, subject to Yesser GSB / data-sharing agreements.	P1	AI-Strategy (Policy Dependency Locator Agent)
BR-22	Implementation Readiness & Feasibility — the system shall assess institutional capacity, funding availability and cross-dependency readiness at the drafting/pre-enactment stage to predict execution risk and implementation feasibility, distinct from post-implementation tracking (CAP-11). Institutional-capacity inputs provided by MOMAH; calibrated in SDD, delivered in Beta.	P2	AI-Strategy (Implementation Risk & Readiness Evaluator)

Note: **P0** Must have, **P1** Should have, **P2** Nice to have, **P3** Backlog

4.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	Multi-factor authentication via Nafath; end-to-end encryption (AES-256) for all data at rest and in transit.
NFR-02	Performance	Level-1 simulation results and AI-assisted draft generation shall complete within 60 seconds; higher simulation tiers (Level 2 / Level 3) to be benchmarked in the SDD.
NFR-03	Scalability	The platform shall scale to process billions of data points as the Policy Knowledge Graph grows.

NFR-04	Language	Full Arabic (RTL) and English support across all user-facing interfaces, policy drafts, and generated outputs.
NFR-05	Explainable AI (XAI) Framework	Every AI-generated clause or simulation outcome shall provide a reasoning path citing the specific data points, laws, or historical precedents used.
NFR-06	Sovereign Compliance	All AI inference and data storage shall remain within the Kingdom, with 100% adherence to NCA and SDAIA AI-ethics and data-residency guidelines (PDPL-compliant).

5. Success Metrics

5.1 Strategic Metrics

Metric Type	Metric	Base-line	Target (6M / Y1 / Y2)	Measurement Method	Fre-quency	KPI Owner
User Adoption	Number of users actively using the Policy AI Lab platform	0	10 / 20 / 50	Platform login logs / active session tracking	Daily	Business Owner / Product Manager
Platform Adoption	Number of policies generated and simulated through the platform	0	3 / 5 / 10 (per BRD Strategic and Operational Metrics)	Simulator Review Screen / policy registry count	Quarterly	Abdulmalik Alshabanat
Revenue Generation	<i>[e.g., SAR revenue increase]</i>	<i>[e.g., 15 M SAR]</i>	<i>[e.g., 25 M SAR]</i>	<i>[How measured and by whom]</i>	<i>[e.g., Quarterly]</i>	<i>Business Owner / Abdulmalik Alshabanat</i>
Cost Savings	<i>[e.g., SAR cost savings]</i>	<i>[e.g., 10 M SAR]</i>	<i>[e.g., 15 M SAR]</i>	<i>[How measured and by whom]</i>	<i>[e.g., Quarterly]</i>	<i>Business Owner / Abdulmalik Alshabanat</i>
Productivity Enhancement	<i>[e.g., SAR value of on time saved]</i>	<i>[e.g., 10 M]</i>	<i>[e.g., 15 M SAR]</i>	<i>[How measured and by whom]</i>	<i>[e.g., Quarterly]</i>	<i>Business Owner / Abdulmalik Alshabanat</i>

Value						
Strategic Alignment	Mean Vision 2030 alignment score of approved policies (0–100)	N/A	TBD — to be set in SDD KPI workshop	Mean Vision 2030 Alignment Score (full HLSD §3.8.2 framework) across approved policies	Quarterly	Abdulmalik Alshabanat
Risk Reduction	Regulatory escalations post-launch (count; lower is better)	TBD (baseline at go-live)	TBD — to be set in SDD KPI workshop	Count of Tier-1/2/3 risk escalations (full HLSD §8.2 Risk Escalation Framework) within 12 months post-enactment	Annual	GRC / Business Owner

Target Rationale & Assumptions

Numeric targets for the financial KPIs (cost savings, productivity, revenue generation) are expressed in SAR millions and will be finalized in the SDD KPI workshop once MOMAH provides financial input values (CAPEX, OPEX, baseline labour cost) and the simulation engine is calibrated.

Cost-savings and productivity assumptions are anchored to the operational cycle-time reduction targets in §5.2 (-10% / -30% / -50% by 6M / Y1 / Y2) applied to current policy-development effort and loaded staff cost.

Revenue-generation and fiscal figures follow a discounted-cash-flow financial methodology: NPV using MoF discount-rate inputs; GDP contribution via direct + indirect + induced impact aligned with GASTAT national accounts and MoF input-output tables; validated against MoF data and Vision 2030 dashboards (BRD §4).

The Vision 2030 alignment score uses the full HLSD §3.8.2 scoring framework (0–100); the post-launch regulatory-escalation count uses the full HLSD §8.2 Risk Escalation Framework tiers. Baselines for these lagging indicators will be established at go-live as a Phase-2 deliverable.

Numeric thresholds for technical KPIs remain pending SDD, as noted in §5.3.

KPI–Business Outcome Linkage

Business Outcome (§2.4)	Linked KPI(s)	Metric section
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Policy Quality Improvement	Mean Vision 2030 alignment score of approved policies; conflict detection rate; assumption documentation completeness	§5.1 / §5.2
Risk Reduction	Regulatory escalations post-launch; risk threshold alert coverage	§5.1 / §5.2
Accelerated Reform Cycles	Policy development cycle-time reduction; policies generated and simulated	§5.2 / §5.1
Financial / business value (business case)	Cost savings; revenue generation; productivity enhancement	§5.1
Adoption	Active platform users	§5.1

5.2 Operational Metrics

Metric Type	Metric	Base line	Target (6M / Y1 / Y2)	Measurement Method	Frequency	KPI Owner
Processes Efficiency	Average time to move a policy from initial concept to approved draft using the AI platform	0	-10% / -30% / -50%	Timestamps at each stage of policy drafting lifecycle	Quarterly	Abdulmalik Alshabanat
Accuracy	Policy simulation accuracy (proportion of simulation outputs accepted without edit in HITL review)	0	TBD / TBD / 80% (per BRD Strategic and Operational Metrics)	HITL approvals without edits / simulation review screen logs; formal metric definition and threshold to be confirmed with MOMAH data owners	Quarterly	Abdulmalik Alshabanat
Quality	Policy conflict	0	TBD / TBD / ≥85% (per	HITL approvals	Quarterly	Abdulmalik

	detection rate — % of actual conflicts flagged by system prior to ministerial review		BRD Strategic and Operational Metrics	cross-referenced with conflict detection logs; ground truth established through: (1) legal expert validation by MOMAH Legal Officers; (2) cross-referencing Maarifa and Sanad platforms; (3) known historical policy contradictions; (4) labelled benchmark dataset to be created during system go-live as a Phase 2 deliverable owned by MOMAH Legal Experts	rly	Alshabanat
Quality	Assumption documentation completeness — % of policies with structured assumption logs in the registry	0	TBD / TBD / 90% (per BRD Strategic and Operational Metrics)	Policy registry audit — % of approved policies with complete assumption records; threshold to be confirmed	Quarterly	Abdulmalik Alshabanat

				after metadata schema is finalized		
Accuracy	Risk threshold alert coverage — % of policies exceeding fiscal/legal risk thresholds for which automated alerts were generated	0	TBD / TBD / >90% (per BRD Strategic and Operational Metrics)	Number of alerts per policy / risk scoring dashboard; coverage target to be confirmed once fiscal/legal thresholds and ground truth are defined	Quarterly	Abdulmalik Alshabanat

5.3 Technical Metrics

Metric Type	Metric	Base-line	Target (6M / Y1 / Y2)	Measurement Method	Frequency	KPI Owner
AI Model Simulation	ABM simulation accuracy vs. historical outcomes	N/A	Target TBD in SDD after data assessment	Back-testing against available GASTAT/SAM A/MoF historical data where sufficient; statistical accuracy measured using an agreed metric such as MAPE, RMSE, or R ² ; formal threshold depends on historical depth and data	Monthly	Abdulmalik Alshabanat

				quality		
AI Model Conflict Detection	Precision/recall of regulatory conflict identification	N/A	Target TBD in SDD after labelled dataset creation	Expert review on labelled conflict sample set; recall is the primary metric, with precision tracked to manage false positives	Monthly	Abdulmalik Alshabanat
AI Model Assumption Extraction	Accuracy of policy assumption extraction from text	N/A	Target TBD in SDD after metadata schema confirmation	Human annotation comparison; NLP evaluation pipeline; threshold depends on field definitions, controlled vocabulary, and document quality	Monthly	Abdulmalik Alshabanat
AI Model Risk Alert Coverage	Coverage of genuine risks flagged by alert system	N/A	Target TBD in SDD after risk taxonomy confirmation	Retrospective comparison against expert-identified risks; threshold depends on agreed risk taxonomy, fiscal/legal trigger rules, and labelled examples	Monthly	Abdulmalik Alshabanat
System Availability	Platform uptime during KSA business hours (07:00–22:00)	N/A	≥99.5% / ≥99.5% / ≥99.5%	Infrastructure APM monitoring dashboard	Weekly	MOMAH IT / DevOps

	AST)					
System Response Time	API response latency (95th percentile)	N/A	≤3s / ≤3s / ≤3s	APM tool; load test reports	Weekly	MOMAH IT / DevOps
System Error Rate	Application-level error rate (5xx HTTP responses)	N/A	<1% / <1% / <1%	APM tool error log analysis	Weekly	MOMAH IT / DevOps
System Draft Generation	Time to generate initial policy draft (95th percentile)	N/A	≤60s / ≤60s / ≤60s	Application performance logs; UAT timing records	Weekly	Policy Officers / MOMAH IT

Note: System-level targets (availability, response time, error rate, draft generation) are indicative and to be confirmed in the SDD/UAT.

6. Roadmap & Phasing

6.1 High-Level Delivery Roadmap

Phase	Key Milestone	Est. Start	Est. End
Preparation	Vision, objectives, scope and requirements defined; BRD approved	—	08 Jun 2026
Prototype	SO, AI Champions and OT approve the business problem, business outcome, success metrics, personas, current landscape, workflows, user journeys and delivery roadmap (Prototype HLSD components)	09 Jun 2026	10 Jul 2026
Alpha	SO, AI Champions and OT approve the business case, integrations,	11 Jul 2026	10 Aug 2026

	architecture, AI approach, data sources, MVP scope and dependencies (Alpha HLSD components)		
Beta	SH and Central AI Committee sign off on the full HLSD; SO, AI Champion and OT validated QAR; Cybersecurity signed-off (SLTO); first go-live	11 Aug 2026	30 Sep 2026
Releases & Stabilization	Go-Live Release; feature and model improvements delivered while stabilizing the product	01 Oct 2026	31 Oct 2026
Operate	OT-validated Transition Support & Maintenance Plan (TSMP) and Knowledge Transfer Compendium (KTP); commonly aligned SDD	01 Nov 2026	Ongoing

Additional resources

Resources to be attached in the appendix by the vendor when completing the Prototype HLSD.

ID	Document type	Description	Link / Attachment Name
DOC-01	Product requirements	Complete list of business and technical requirements for the solution.	Policy AI Lab - Product Requirements Document (PRD) v1.0.pdf
DOC-02	Solution to services mapping	Solution mapping to the existing MOMAH services. Validate and update the mapping of this solution to the respective MOMAH services that the solution directly provides.	HLSD - MOMAH Services Mapping.xlsx

Appendix

A. References

- SDAIA National AI Ethics Principles (2025)
- NIST AI Risk Management Framework (AI RMF 1.0)
- Saudi Personal Data Protection Law (PDPL)
- NHCI AI Programme Governance Framework
- Recommendations on AI Evaluation
- Policy_AI_Lab_BRD_v2.0.docx

B. Approval Sign-Off

Role	Name	Signature	Date
Solution Vendor (Prepared By)			
Solution Owner (Reviewed By)			
NHCI Oversight (Reviewed By)			
Central AI (Reviewed By)			
AI Champion (Reviewed By)			
Sector Head (Reviewed By)			
AI Solutions Steering Group (Approved By)			
Central AI Committee (Approved By)			